

Contents

Volume 02 — Advanced VAEs and Generative Adversarial Networks	1
L10: Conditional VAE (CVAE)	1
L11: Beta-VAE and Disentangled Representations	2
L12: Latent Space Interpolation and VAE Limitations	2
L13: Generative Adversarial Networks: Minimax Formulation	3
L14: GAN Training Dynamics and Mode Collapse	4
L15: DCGAN Architecture and Design Principles	5
L16: Conditional GAN and Pix2Pix	5
L17: CycleGAN and Unpaired Image Translation	6
L18: StyleGAN and GAN Evaluation Metrics	6
Volume 02 summary	7

Volume 02 — Advanced VAEs and Generative Adversarial Networks

Lectures L10-L18 · Weeks 4-6

L10: Conditional VAE (CVAE)

Learning objectives

- Extend VAEs to conditional generation $p(x | c)$.
- Derive the conditional ELBO.

Conditional generative model

Given condition c (class label, text caption, attribute vector):

$$p_{\theta}(x | c) = \int p_{\theta}(x | z, c) p(z | c) dz$$

The encoder becomes $q_{\phi}(z | x, c)$ and the decoder $p_{\theta}(x | z, c)$.

Conditional ELBO

$$\mathcal{L}(\theta, \phi; x, c) = \mathbb{E}_{q_{\phi}(z|x,c)}[\log p_{\theta}(x | z, c)] - D_{\text{KL}}(q_{\phi}(z | x, c) \| p(z | c))$$

Typically $p(z | c) = p(z) = \mathcal{N}(0, I)$.

Applications

- Class-conditional image generation (MNIST digits by label)
 - Text-to-image conditioning (caption as c)
 - Controlled attribute manipulation
-

L11: Beta-VAE and Disentangled Representations

Learning objectives

- Understand disentanglement in latent spaces.
- Formulate the β -VAE objective.

Disentanglement

A representation is **disentangled** if each latent dimension controls a single generative factor (e.g., rotation, color, scale) independently.

β -VAE objective

$$\mathcal{L}_{\beta\text{-VAE}} = \mathbb{E}_{q_\phi} [\log p_\theta(x | z)] - \beta \cdot D_{\text{KL}}(q_\phi(z | x) \| p(z))$$

With $\beta > 1$, stronger pressure on the KL term encourages **factorized** latent codes at the cost of reconstruction fidelity.

Evaluation metrics

- **MIG** (Mutual Information Gap)
 - **SAP** (Separated Attribute Predictability)
 - **DCI** (Disentanglement, Completeness, Informativeness)
-

L12: Latent Space Interpolation and VAE Limitations

Learning objectives

- Perform latent space interpolation and arithmetic.
- Identify fundamental VAE limitations.

Latent interpolation

Given encodings $z_1 = \mu_\phi(x_1)$ and $z_2 = \mu_\phi(x_2)$:

$$z(\alpha) = (1 - \alpha) z_1 + \alpha z_2, \quad \alpha \in [0, 1]$$

Decode $\hat{x}(\alpha) = \mu_\theta(z(\alpha))$ for smooth morphing between inputs.

Latent arithmetic

$$z_{\text{result}} = z_{\text{king}} - z_{\text{man}} + z_{\text{woman}}$$

VAE limitations

Limitation	Cause
Blurry outputs	Gaussian decoder assumption; MSE loss
Posterior collapse	$\text{KL} \rightarrow 0$; decoder ignores z
Weak samples	ELBO is loose; $p_{\theta}(x)$ underestimated

These motivate adversarial training (GANs, next section).

L13: Generative Adversarial Networks: Minimax Formulation

Learning objectives

- Formulate GAN training as a two-player game.
- Derive the minimax objective and its practical variant.

GAN architecture

flowchart LR

```
Z["z ~ p(z)"] --> G["Generator G_θ"]
G --> FAKE["Fake sample G(z)"]
REAL["Real sample x"] --> D["Discriminator D_φ"]
FAKE --> D
D --> OUT["Real / Fake probability"]
OUT -->| "Train G: fool D" | G
OUT -->| "Train D: classify" | D
```

- **Generator** $G_{\theta} : \mathcal{Z} \rightarrow \mathcal{X}$ maps noise to data.
- **Discriminator** $D_{\phi} : \mathcal{X} \rightarrow [0, 1]$ classifies real vs. fake.

Minimax objective

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$

Optimal discriminator

For fixed G , the optimal discriminator is:

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)}$$

Global optimum

At equilibrium, $p_G = p_{\text{data}}$ and $D^*(x) = \frac{1}{2}$.

Non-saturating generator loss

In practice, the generator minimizes:

$$\mathcal{L}_G = -\mathbb{E}_{z \sim p(z)} [\log D(G(z))]$$

This avoids vanishing gradients when D confidently rejects fakes.

Key equation summary

$$\min_{\theta} \max_{\phi} \mathbb{E}_x [\log D_{\phi}(x)] + \mathbb{E}_z [\log(1 - D_{\phi}(G_{\theta}(z)))]$$

L14: GAN Training Dynamics and Mode Collapse

Learning objectives

- Analyze GAN training instability.
- Describe mode collapse and mitigation strategies.

Training challenges

Problem	Description
Mode collapse	Generator produces limited variety
Vanishing gradients	D too strong; G receives no useful signal
Oscillation	No stable Nash equilibrium
Non-convergence	Alternating updates don't guarantee convergence

Mode collapse

The generator maps many z values to a few outputs, failing to cover the full data distribution. Detection: low diversity in generated samples; low inception score variance.

Mitigation techniques

1. **Minibatch discrimination** — discriminator sees batch statistics.
2. **Unrolled GANs** — simulate future discriminator updates.
3. **Spectral normalization** — constrain discriminator Lipschitz constant.
4. **Wasserstein GAN (WGAN)** — replace JS divergence with Wasserstein distance:

$$\mathcal{L}_{\text{WGAN}} = \mathbb{E}_{x \sim p_{\text{data}}} [D(x)] - \mathbb{E}_{z \sim p(z)} [D(G(z))]$$

with weight clipping or gradient penalty on D .

L15: DCGAN Architecture and Design Principles

Learning objectives

- Apply DCGAN architectural guidelines.
- Understand transposed convolutions for upsampling.

DCGAN guidelines (Radford et al., 2016)

1. Replace pooling with strided convolutions (discriminator) and transposed convolutions (generator).
2. Use batch normalization in both G and D .
3. Remove fully connected hidden layers; use global average pooling.
4. Use ReLU in G (except output: tanh); LeakyReLU (0.2) in D .

Generator structure

z (100-d) $\rightarrow$ FC $\rightarrow$ 4×4×512 $\rightarrow$ ConvT $\rightarrow$ 8×8×256 $\rightarrow \dots \rightarrow$ 64×64×3

Each transposed convolution doubles spatial resolution while halving channels.

Transposed convolution

For upsampling, transposed convolution (deconvolution) learns the upsampling kernel:

$$y_{i,j} = \sum_{m,n} W_{m,n} \cdot x_{\lfloor i/s \rfloor, \lfloor j/s \rfloor}$$

L16: Conditional GAN and Pix2Pix

Learning objectives

- Condition GAN generation on labels or paired inputs.
- Describe the Pix2Pix image-to-image framework.

Conditional GAN (cGAN)

$$\min_G \max_D \mathbb{E}_{x,y} [\log D(x,y)] + \mathbb{E}_{z,y} [\log(1 - D(G(z,y), y))]$$

Condition y is concatenated to input (channel-wise) or embedded and added.

Pix2Pix

For paired image translation (e.g., sketch $\rightarrow$ photo):

$$\mathcal{L} = \mathcal{L}_{\text{cGAN}} + \lambda \mathcal{L}_{\text{L1}}$$

where $\mathcal{L}_{\text{L1}} = \|y - G(x)\|_1$ enforces pixel-level fidelity.

Uses a **U-Net generator** with skip connections and a **PatchGAN discriminator** that classifies local $N \times N$ patches.

L17: CycleGAN and Unpaired Image Translation

Learning objectives

- Translate between domains without paired training data.
- Explain cycle consistency loss.

Problem setting

Domains X and Y with unpaired samples. Learn $G : X \rightarrow Y$ and $F : Y \rightarrow X$.

Cycle consistency

$$\mathcal{L}_{\text{cycle}} = \mathbb{E}_{x \sim p_X} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_Y} [\|G(F(y)) - y\|_1]$$

Full objective

$$\mathcal{L} = \mathcal{L}_{\text{GAN}}(G, D_Y) + \mathcal{L}_{\text{GAN}}(F, D_X) + \lambda \mathcal{L}_{\text{cycle}}$$

Applications

- Horse $\leftrightarrow$ Zebra, summer $\leftrightarrow$ winter, photo $\leftrightarrow$ Monet painting
-

L18: StyleGAN and GAN Evaluation Metrics

Learning objectives

- Understand StyleGAN's style-based generator.
- Apply FID and related evaluation metrics.

StyleGAN innovations

1. **Mapping network** — transforms z to intermediate latent w in a less entangled space.
2. **Adaptive instance normalization (AdaIN)** — injects style at each layer:

$$\text{AdaIN}(x, y) = y_s \cdot \frac{x - \mu(x)}{\sigma(x)} + y_b$$

3. **Progressive growing** — train on low resolution first, then add layers.

Evaluation metrics

Fréchet Inception Distance (FID):

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

where (μ_r, Σ_r) and (μ_g, Σ_g) are mean and covariance of Inception features for real and generated images. Lower is better.

Inception Score (IS):

$$\text{IS} = \exp(\mathbb{E}_x [D_{\text{KL}}(p(y | x) \| p(y))])$$

Higher IS indicates diverse, classifiable generated images.

Volume 02 summary

Lecture	Core concept
L10	Conditional VAE
L11	β -VAE, disentanglement
L12	Latent interpolation, VAE limits
L13	GAN minimax objective
L14	Mode collapse, WGAN
L15	DCGAN architecture
L16	cGAN, Pix2Pix
L17	CycleGAN
L18	StyleGAN, FID/IS

Previous: [Volume 01](#) · **Next:** [Volume 03 — Diffusion and Transformers](#)